Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 3 **Winter Semester 2024-2025**

DURATION: 30 MIN **FULL MARKS: 20**

CSE 4107: Structured Programming I

Figures in the right margin indicate marks.

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1. Find the output of the following program: **5 (CO1) (PO2)**

```
#include <stdio.h>
int main() {
    int arr[6] = {2, 4, 6, 8, 10, 12};
    int i, sum = 0;

    for (i = 0; i < 6; i++) {
        if (i % 2 == 0)
            arr[i] = arr[i] + i;
        else
            arr[i] = arr[i] - i;
        sum = sum + arr[i];
    }
    printf("Sum = %d\n", sum);

    for (i = 5; i >= 0; i--) {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

2. Write the full definition of a function declared as follows: **10 (CO3) (PO1)**

`int second_largest(int arr[], int n)`

This function will output the second largest **distinct** element in that array. In your main function, scan a positive integer, `n`, from the user and then scan `n` number of integers into an array. Pass this array and `n` into the `second_largest` function to generate the output. A sample input-output scenario is given below:

```
Enter n: 6
Enter 6 integers: 3 12 7 12 9 4

Second Largest: 9
```

3. Find the output of the following program: **5 (CO1) (PO2)**

```
#include <stdio.h>

int x = 5;

int modify(int x) {
    x = x + 3;
    return x;
}

int update(int y) {
    y = y * 2;
    x = x + y;
    return y;
}

int main() {
    int x = 10;
    int a, b;
    a = modify(x);
    b = update(a);
    printf("x = %d\n", x);
    printf("a = %d\n", a);
    printf("b = %d\n", b);
    return 0;
}
```